

Gin Theih Tuang

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EDUCATION

Northwestern University | Evanston, Illinois

Graduation Date: June 2028

Bachelor of Science in Mechanical Engineering

Relevant Coursework: Thermodynamics, Fluid Mechanics, Heat Transfer, Stress Analysis, Design and Manufacturing, Theory of Machines and Dynamics, Material Science, and Mechanics of Material.

SKILLS

- Design and Analysis: SolidWorks (CAD), CFD (Computational Fluid Dynamics), Topology Optimization, Finite Element Analysis (FEA), Rapid Prototyping.
- Computation: MATLAB (Numerical Modeling), Python, Microsoft Excel.
- Interpersonal: Articulate Communication, Collaborative Teamwork, Inquisitive Learner, Prompt Responsiveness.
- Languages: English, Zopau (Official Oklahoma Seal of Bilingualism).

PROJECT EXPERIENCE

Formula SAE (Northwestern Formula Racing) | Evanston, Illinois

September 2025 - Present

Aero Sub-team Member

- Engineered structural rear wing inserts using SolidWorks, utilizing complex geometry to ensure seamless integration between the carbon fiber skin and internal mounting points.
- Executed Topology Optimization and FEA on aluminum mounts and inserts, achieving a 33% mass reduction (from 9 lbs to 6 lbs) while maintaining a Factor of Safety (FoS) of 2.0+ under peak aerodynamic loading.
- Conducted CFD Simulations on the rear wing assembly to optimize the Angle of Attack (AoA) for the secondary flap, maximizing downforce while maintaining an efficient lift-to-drag ratio.

Engineering Analysis 4 (Simulation Projects) | Evanston, Illinois

September 2025 - December 2025

Undergraduate Researcher (Computational Modeling)

- Engineered custom adaptive-step ODE solvers in MATLAB (Improved Euler) to simulate stiff nonlinear systems, achieving a 3.76x reduction in computational cost while maintaining a global error tolerance of 10^{-7} .
- Modeled complex system behaviors including limit cycle attractors, bifurcation points, and chaotic transients; validated numerical results against analytical benchmarks, achieving <1% discrepancy in high-stiffness regimes.
- Quantified simulation reliability by analyzing "trust windows" across 10 orders of magnitude of error tolerance (10^{-3} to 10^{-13}) and utilized linear interpolation to extract high-fidelity period data from non-uniform datasets.

Design Thinking and Communication (DTC) | Evanston, Illinois

September 2024 - June 2025

Undergraduate Design Engineer

- Collaborated on a multidisciplinary team to design and prototype an ergonomic pediatric oxygen tank holder; utilized user-research to improve safety factors and transportability for high-mobility patients.
- Directed a campus transit vehicle prototype build and managed technical data via a project website, ensuring adherence to design requirements and increasing project visibility.

WORK EXPERIENCE

Barnes & Noble College (Northwestern University) | Evanston, Illinois

November 2025 - Present

Student Associate

- Streamlined inventory flow by labeling, sorting, and restocking 50+ apparel and textbook items per shift, ensuring 100% store readiness for high-volume student traffic.
- Managed point-of-sale (POS) systems and processed 10+ digital textbook orders per shift with zero-error scanning, maintaining high efficiency while multitasking backroom logistics.

COMMUNITY LEADERSHIP

- Co-Organizer, TEDx Youth @ Jenks: Managed logistics for 300+ attendees and 20 speakers in Tulsa, OK.
- Youth Camp Counselor, FGA Int'l: Directed 15+ youth sessions and community programs for 300+ participants.
- Member: American Society of Mechanical Engineers (ASME), Formula SAE.